

Solution to Exercise 04 — Analysis on Manifolds, 80416

May 13, 2026



Question 1

Subquestion 1

For $a, b > 0$ Let $\mathcal{E}_{a,b} := \left\{ (x, y) \in \mathbb{R}^2 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 \leq 1 \right\}$ be the ellipse with parameters a, b .

We will use Green's theorem to show $\text{Area}(\mathcal{E}_{a,b}) = \pi ab$.

Solution: First we notice that $\partial\mathcal{E}_{a,b} = \left\{ (x, y) \in \mathbb{R}^2 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1 \right\}$ and ellipse is a smooth domain.

We need a positive orientated parameterization which is given by $\gamma(t) = (a \cos(t), b \sin(t))$ for $t \in [0, 2\pi]$.

We denote $x(t) = a \cos(t)$, $y(t) = b \sin(t)$ hence

$$dx = x'(t) dt = -a \sin(t) dt$$

$$dy = y'(t) dt = b \cos(t) dt$$

In the recitation we saw the corollary

$$\text{Area}(\mathcal{E}_{a,b}) = \int_{\partial\mathcal{E}_{a,b}} x dy = - \int_{\partial\mathcal{E}_{a,b}} y dx = \frac{1}{2} \int_{\partial\mathcal{E}_{a,b}} x dy - y dx$$

So in our case

$$\text{Area}(\mathcal{E}_{a,b}) = \int_{\partial\mathcal{E}_{a,b}} x dy = \int_0^{2\pi} a \cos(t) \cdot b \cos(t) dt = ab \int_0^{2\pi} \cos^2(t) dt = ab\pi$$

Where the last equals is from the known identity $\int_0^{2\pi} \cos^2(t) dt = \pi$. □

Subquestion 2

Let $\Omega = \left\{ (x, y) \in \mathbb{R}^2 \mid x^{\frac{2}{3}} + y^{\frac{2}{3}} \leq 1 \right\}$ and we will compute the perimeter of Ω .

Solution: This is another ellipse so the parameterization is $\gamma(t) = (\cos^3(t), \sin^3(t))$ for $t \in [0, 2\pi]$ which is smooth and we can see that

$$(\cos^3(t))^{\frac{2}{3}} + (\sin^3(t))^{\frac{2}{3}} = \cos^2(t) + \sin^2(t) = 1$$

We differentiate

$$x'(t) = -3 \cos^2(t) \sin(t)$$

$$y'(t) = 3 \sin^2(t) \cos(t)$$

Hence

$$\|\dot{\gamma}(t)\| = \sqrt{x'(t)^2 + y'(t)^2} = \sqrt{9 \cos^4(t) \sin^2(t) + 9 \sin^4(t) \cos^2(t)} = 3 \sqrt{\cos^2(t) \sin^2(t) (\cos^2(t) + \sin^2(t))} = 3 |\sin(t) \cos(t)|$$

The perimeter is given by

$$L = \int_0^{2\pi} \|\dot{\gamma}(t)\| dt = 3 \int_0^{2\pi} |\sin(t) \cos(t)| dt = 12 \int_0^{\frac{\pi}{2}} \sin(t) \cos(t) dt = 12 \cdot \left[\frac{1}{2} \sin^2(t) \right]_{t=0}^{t=\frac{\pi}{2}} = 12 \cdot \frac{1}{2} = 6$$

□

Subquestion 3

Let $P \subseteq \mathbb{R}^2$ be a polygon whose vertices in a counterclockwise order are $\{p_1, \dots, p_n\} \subseteq \mathbb{R}^2$.

For every $1 \leq i \leq n$ denote $p_i = (x_i, y_i)$ and we will prove that $\text{Area}(P) = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i)$ where $x_{n+1} = x_1$, $y_{n+1} = y_1$.

Solution: First we notice that P a polygon is a simply connected domain and from the previous exercise in a simply connected domain, the property that all curves with the same endpoints are homotopic ensures that locally conservative fields become conservative.

We need to define a vector field $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and by the identities we saw in the recitation we choose

$$F(x, y) = \frac{1}{2} \begin{pmatrix} -y \\ x \end{pmatrix}$$

Since we need to treat x, y as equals this is the most fitting function from the properties in the recitation.

So by Green's Theorem and the recitation

$$\text{Area}(P) = \oint_{\partial P} F \, d\ell = \int_{\partial P} \frac{1}{2}(x \, dy - y \, dx)$$

The boundary ∂P can be represented as a piecewise C^1 path so we define this boundary as the union of n directed (counterclockwise) line segments

$$\partial P = \bigcup_{i=1}^n \gamma_i$$

Where each γ_i is the parametrized curve starting at $p_i = (x_i, y_i)$ and ends at $p_{i+1} = (x_{i+1}, y_{i+1})$ where $p_{n+1} = p_1$. Therefore each $\gamma_i : [0, 1] \rightarrow \mathbb{R}^2$ is given by

$$\gamma_i(t) = tp_{i+1} + (1-t)p_i = \begin{pmatrix} x_i + t(x_{i+1} - x_i) \\ y_i + t(y_{i+1} - y_i) \end{pmatrix}$$

We have

$$\dot{\gamma}_i(t) = \begin{pmatrix} x_{i+1} - x_i \\ y_{i+1} - y_i \end{pmatrix} = \begin{pmatrix} \Delta x_i \\ \Delta y_i \end{pmatrix}$$

By the properties of the line integral we have

$$\text{Area}(P) = \sum_{i=1}^n \int_{\gamma_i} F \, d\ell$$

We compute

$$\begin{aligned} \int_{\gamma_i} F \, d\ell &= \int_0^1 \langle F(\gamma_i(t)), \dot{\gamma}_i(t) \rangle \, dt \\ &= \frac{1}{2} \int_0^1 (-y_i + t\Delta y_i) \Delta x_i + (x_i + t\Delta x_i) \Delta y_i \, dt \\ &= \frac{1}{2} \int_0^1 -y_i \Delta x_i - t\Delta y_i \Delta x_i + x_i \Delta y_i + t\Delta x_i \Delta y_i \, dt \\ &= \frac{1}{2} \int_0^1 x_i \Delta y_i - y_i \Delta x_i \, dt \\ &= \frac{1}{2} (x_i \Delta y_i - y_i \Delta x_i) \end{aligned}$$

We substitute back $\Delta x_i = x_{i+1} - x_i$ and $\Delta y_i = y_{i+1} - y_i$ and get that

$$\int_{\gamma_i} F \, d\ell = \frac{1}{2} (x_i(y_{i+1} - y_i) - y_i(x_{i+1} - x_i)) = \frac{1}{2} (x_i y_{i+1} - x_{i+1} y_i)$$

Hence

$$\text{Area}(P) = \sum_{i=1}^n \int_{\gamma_i} F \, d\ell = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i)$$

□

Question 2

Assume that $A = \pi R^2$.

Subquestion 1

We will prove that $L = 2\pi r$.

Proof: L is the perimeter of the circle and from the isoperimetric inequality from the recitation we can deduce $A \leq \frac{L^2}{4\pi}$ and equality holds for balls.

But a disk is a ball hence $A = \frac{L^2}{4\pi}$ but $A = \pi r^2$ so $\pi r^2 = \frac{L^2}{4\pi}$ which means $4\pi^2 r^2 = L^2$ and since $L > 0$ we can take square roots and have $L = 2\pi r$. $\square$

Subquestion 2

Denote $\alpha' = \min_{t \in [0, T_1]} y(t)$ and $\beta' = \max_{t \in [0, T_1]} y(t)$ and we will show that $r = \frac{\beta' - \alpha'}{2}$.

Proof: In the recitation we set $\alpha = \min_{t \in [0, T_1]} x(t)$ and $\beta = \max_{t \in [0, T_1]} x(t)$.

We assume that the curve parametrizes a circle of radius r centered at (x_0, y_0) then

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

And for every point on the circle

$$x_0 - r \leq x \leq x_0 + r$$

Moreover the points $(x_0 \pm r, y_0)$ are in the circle hence $\alpha = x_0 - r$ and $\beta = x_0 + r$.

Therefore

$$\beta - \alpha = (x_0 + r) - (x_0 - r) = 2r \iff r = \frac{\beta - \alpha}{2}$$

Similarly

$$y_0 - r \leq y \leq y_0 + r$$

And the point $(x_0, y_0 \pm r)$ are in the circle hence $\alpha' = y_0 - r$ and $\beta' = y_0 + r$ thus

$$\beta' - \alpha' = (y_0 + r) - (y_0 - r) = 2r \iff r = \frac{\beta' - \alpha'}{2}$$

$\square$

Subquestion 3

We will prove that $x(t) = \pm r \dot{y}(t)$ and $y(t) = \pm r \dot{x}(t)$ then we will deduce that $\Omega = B_\Omega$ (we will assume here that the parametrization of γ is by arc-length parameterization).

Proof: In the recitation we saw that

$$A + \pi r^2 = \int_0^{T_1} \left\langle \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \begin{pmatrix} y'(t) \\ -x'(t) \end{pmatrix} \right\rangle dt \leq \int_0^{T_1} \left\| \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} \right\| \cdot \left\| \begin{pmatrix} y'(t) \\ -x'(t) \end{pmatrix} \right\| dt$$

We can see that equality in the isoperimetric inequality means equality holds in Cauchy-Schwarz, therefore the two vectors are linearly dependent for every t which means that there exists $\lambda(t)$ such that

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \lambda(t) \begin{pmatrix} y'(t) \\ -x'(t) \end{pmatrix} \iff x(t) = \lambda(t)y'(t) \quad y(t) = -\lambda(t)x'(t)$$

From the assumption the parametrization is by arc length which means

$$x'(t)^2 + y'(t)^2 = 1$$

From the construction of the ball B we also have

$$x(t)^2 + y(t)^2 = r^2$$

Therefore

$$r^2 = x(t)^2 + y(t)^2 = \lambda(t)^2 y'(t)^2 + \lambda(t)^2 x'(t)^2 = \lambda(t)^2 (y'(t)^2 + x'(t)^2) = \lambda(t)^2 \implies |\lambda(t)| = r$$

But $\lambda(t)$ is continuous and $\lambda(t) \neq 0$ for all t so it's sign is a constant thus either $\lambda(t) = r$ for all t or $\lambda(t) = -r$ for all t which exactly means $x(t) = \pm r\dot{y}(t)$ and $y(t) = \pm r\dot{x}(t)$.

We saw that

$$x(t)^2 + y(t)^2 = r^2(\dot{x}(t)^2 + \dot{y}(t)^2)$$

But since $\dot{x}(t)^2 + \dot{y}(t)^2 = 1$ we have $x(t)^2 + y(t)^2 = r^2$.

Therefore the image of the curve lies on the circle of radius r so $\partial\Omega = \partial B_r(0)$ which are both bounded domains with the same boundary so we can conclude that $\Omega = B_r(0)$. $\square$

Question 3

Let $\Omega \subseteq \mathbb{R}^2$ be a domain. Let $p \in \partial\Omega$ and let $p \in V \subseteq \mathbb{R}^2$ be a neighborhood of p .

Denote by $B = B_1(0) \subseteq \mathbb{R}^2$ the unit ball centered at 0 and we will assume that $\varphi : B \rightarrow V$ is a diffeomorphism.

We also assume that $\varphi(\{y > 0\}) = V \cap \Omega$ and $\varphi(\{y = 0\}) \subseteq \partial\Omega$.

We will prove that $\det d\varphi > 0$ if and only if the curve $t \mapsto \varphi(t, 0)$ is a positively oriented with respect to Ω .

Formally, we say that the boundary of a domain $\Omega \subseteq \mathbb{R}^2$ parametrized by $\gamma : I \rightarrow \partial\Omega$, $\gamma(t) = (x(t), y(t))$ is positively oriented if

$$\forall t \in I, \exists \delta > 0, \forall h \in (0, \delta), \gamma(t) + h\hat{n}(t) \in \Omega$$

Where $\hat{n}(t) = (-\dot{y}(t), \dot{x}(t))$.

Proof: First we need to understand the geometric definition of φ .

One can see that on the x -axis (where $y = 0$) we go to the boundary so φ maps the x -axis of the ball into the boundary of Ω .

On the upper half ball, where $y > 0$, we go to the “inside” of Ω which means the φ maps the ball into the interior of Ω .

If we look at the vector $e_2 = (0, 1)$ this vector points from the boundary into the interior of the ball which means that its image under $d\varphi(e_2)$ must therefore points into the domain Ω .

If we set the coordinates in the ball to be (u, v) and the coordinates in Ω to be (x, y) , the Jacobian matrix $d\varphi$ is

$$\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

Hence

$$\det d\varphi = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} = \dot{x} \frac{\partial y}{\partial v} - \dot{y} \frac{\partial x}{\partial v}$$

We define $\gamma(t) = \varphi(t, 0)$ so $\dot{\gamma}(t) = \frac{\partial \varphi}{\partial u}(t, 0) = \begin{pmatrix} \dot{x}(t) \\ \dot{y}(t) \end{pmatrix}$ this is the boundary curve and we notice that $\hat{n}(t)$ is a counterclockwise rotation of $\dot{\gamma}(t)$.

Since $\varphi(\{v > 0\}) \subseteq \Omega$ the vector $\frac{\partial \varphi}{\partial v}$ points into the domain Ω .

We compute the dot product between $\hat{n}(t)$ and the inward pointing vector $v_2 = \frac{\partial \varphi}{\partial v}$

$$\left\langle \begin{pmatrix} -\dot{y} \\ \dot{x} \end{pmatrix}, v_2 \right\rangle = \left\langle \begin{pmatrix} -\dot{y} \\ \dot{x} \end{pmatrix}, \begin{pmatrix} \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial v} \end{pmatrix} \right\rangle = -\dot{y} \frac{\partial x}{\partial v} + \dot{x} \frac{\partial y}{\partial v} = \det d\varphi$$

$\implies$ If $\det d\varphi > 0$ then the dot product $\langle \hat{n}(t), v_2 \rangle > 0$ which actually means that $\gamma(t)$ is positively oriented since v_2 points into Ω and $\hat{n}(t)$ must also point into Ω .

$\impliedby$ If $\gamma(t)$ is positively oriented then $\hat{n}(t)$ must points into Ω since φ is a diffeomorphism and v_2 points into Ω the dot product must be positive which implies $\det d\varphi > 0$. $\square$

Question 4

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be C^1 functions such that $f \leq g$ and we denote $A = \{(x, y) \in \mathbb{R}^2 \mid x \in [a, b], y \in [f(x), g(x)]\}$.

Let Ω be a neighborhood of A and let $F = (P, Q) : \Omega \rightarrow \mathbb{R}^2$ be a smooth vector field.

Let $R \in M_2(\mathbb{R})$ be an orthogonal matrix such that $\det R = 1$.

Denote $A_R = \{Rp \in \mathbb{R}^2 \mid p \in A\}$ and $\Omega_R = \{Rp \in \mathbb{R}^2 \mid p \in \Omega\}$ and we define the vector field $F_R = (P_R, Q_R) : \Omega_R \rightarrow \mathbb{R}^2$ by $F_R = R \circ F \circ R^{-1}$.

We will prove that

$$\int_{\partial A_R} F_R d\ell = \int_{A_R} \frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} dx dy$$

Proof: We take $\gamma : [c, d] \rightarrow \mathbb{R}^2$ a parameterization of the boundary ∂A .

Since A_R is the image of A under the rotation it can be parametrized by $\sigma(t) = R\gamma(t)$ and $\dot{\sigma}(t) = R\dot{\gamma}(t)$.

The line integral of F_R along ∂A_R is defined as

$$\begin{aligned} \oint_{\partial A_R} F_R d\ell &= \int_c^d F_R(\sigma(t)) \dot{\sigma}(t) dt \\ &= \int_c^d (R \circ F \circ R^{-1})(R\gamma(t)) \cdot (R\dot{\gamma}(t)) dt \\ &= \int_c^d (RF(\gamma(t))) \cdot (R\dot{\gamma}(t)) dt \\ &= \int_c^d F(\gamma(t)) \cdot \dot{\gamma}(t) dt = \oint_{\partial A} F d\ell \end{aligned}$$

Where the second equality is substituting the definition of F_R and $\sigma(t)$ and the forth equality is since R is an orthogonal matrix which means that $\langle Ru, Rv \rangle = \langle u, v \rangle$.

By change of variable we set $(x, y) = R(u, v)$ and since $\det R = 1$ the Jacobian determinant is $|\det R| = 1$ so the double integral becomes

$$\iint_{A_R} \frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} dx dy = \iint_A \left(\frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} \right) |_{R(u,v)} \cdot 1 du dv$$

We need to simplify the integral: if dF is the Jacobian of the original field then the Jacobian of the rotated field F_R is $R dF R^{-1}$.

Furthermore, $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ is rotational invariant since it represents the trace of the field's derivative multiplied by a matrix rotation (from the cyclic property of the trace) so we will have

$$\left(\frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} \right) |_{R(u,v)} = \frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v}$$

Therefore

$$\iint_{A_R} \frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} dx dy = \iint_A \frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v} du dv$$

Applying Green's Theorem gives us

$$\oint_{\partial A} F d\ell = \iint_A \frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v} du dv$$

Hence from what we found

$$\oint_{\partial A} F d\ell = \iint_{A_R} \frac{\partial Q_R}{\partial x} - \frac{\partial P_R}{\partial y} dx dy$$

□